

MADS: Modular Arithmetic and Diophantine Scheduling

Submitted to:

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by

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Abstract

Workplaces often are not unfamiliar to having a huge workload often times brought upon by too many people to accommodate for which is manageable on cases like resorts, hotels, hospitals and even government related offices through the process of making appointments to control the amount of people on a certain time frame, however even with practices like this workplaces are still prone to overwork. These problems emerge more often than people think and have more drawbacks than just wasting time and causing overwork. For this reason the researchers have come up with a method of managing appointments using both Modular Arithmetic and Linear Diophantine Equation. After analysis results have proven to have a positive effect to which the problem has been addressed and as well as other positive byproducts such as a better work environment and service quality.

Introduction

Everyone has experienced a long queue whether it's for an Appointment, Renewing ID and many more. This can easily be cumbersome and time consuming to deal with. In this paper the researchers aim to improve that by being able to distribute the load between many different days through advanced scheduling helping to avoid long lines and promote shorter and more efficient queues. Applying the concept of Modular Arithmetic and Linear Diophantine to find the best available time for the following week and to help allot the appointments to the available week.

Theoretical Framework

A. Modular Arithmetic

In number theory, modular arithmetic looks at both whole numbers and the numbers that are left over when you split them by a certain set of positive integers. This set of integers is called the modulus (Libretexts, 2021). It works especially well when things behave in cycles or repeating patterns, like the days of the week, the months of the year, or plans that happen over and over again. The general expression for modular congruence is:

$$a \equiv b \pmod{m}$$

This denotes that integers a and b yields the same remainder when divided by modulus m .

Application to Scheduling:

In appointment scheduling, modular arithmetic can be utilized to identify recurring time slots within a fixed schedule. It enables the determination of future availability by computing the position within a cycle. The formula is expressed as:

$$Gap = (total\ weeks + 1) - current\ week$$

$$Next\ Week = ((Current\ Week + Gap) \bmod Total\ Weeks) + 1$$

Example:

Assume that appointments recur every 3 weeks and the current week is Week 2. To calculate the week of the next available appointment:

$$((2 + 2) \bmod 3) + 1 = 4 \quad (4 \bmod 3) + 1 = 2$$

Means that the next most preferable appointment would be scheduled within **Week 2 (current week) to week 1 (2 weeks from now** in a 3 week cycle) in other words, **within the next 2 weeks.**

B. Linear Diophantine Equation

Diophantine equations are equations where we're only looking for whole number solutions. These are useful in real-life situations where in our situation we can't deal with fractions. Appointment Scheduling is assigning actual people to time slots. We are dealing with a whole number since the appointment requires a whole appointment.

The general form of a **linear Diophantine equation** is:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = N$$

where x_1, x_2, x_n are unknowns (typically representing quantities to be assigned), and N is a known total.

Example:

The total number of appointments is 30, and we want to split them across 4 weeks. Each variable x_i stands for how many appointments we're assigning to a specific week.

The solution can be:

$$x_1 = 8, x_2 = 7, x_3 = 7, x_4 = 8$$

$$8 + 7 + 7 + 8 = \mathbf{30}$$

Application

Scenario:

According to the most recent survey done in 2018 a physician's saw between 11 and 20 patients per day and spent between 17 and 24 minutes with each patient during the day. (*Rajae, 2022*). With this in mind we can say on average per week a physician's saw between 77 and 140 therefore an effective and efficient schedule should be made to maximize the amount of patients seen by a physician.

But physician's need to avoid burnouts because its causes decrease efficiency in diagnoses of patients therefore risking the overall health of the patient. Working too many hours can lead to burnout among physicians 57% of doctors working 71 or more hours during a week leads to burnout and 50% working 61-70 hours during a week report burnout. Nearly half of physicians suffering from burnout also reported they work 51–60 hours, whereas 36% claimed to work 31-40 hours per week. With this we can say that the ideal hours per week is less than 40 hours and if on average a physician spends 20 minutes per patient, we can say that per week a physician should only have around 120 patients per week.

1 hour = 60 minutes

40 hours = $40 \times 60 = 2400$ minutes/week

Each patient takes 20 minutes, so:

$$\frac{2400 \text{ minutes}}{20 \text{ minutes per patient}} = 120 \text{ patients/week}$$

A clinic has **3 available weeks** for scheduling, **140 patients** need appointments. The Clinic can handle **no more than 120 patients** per cycle. Today is **Week 2**, what is the next available week for scheduling given that the current week and the following are occupied.

Using Modular Arithmetic to cycle weeks:

Using the formula,

$$Gap = (total\ weeks + 1) - current\ week$$

$$Schedule = ((Current\ Week + Gap) \bmod Total\ Weeks) + 1$$

We have:

$$Gap = (total\ 3 + 1) - 2 = 2$$

$$Schedule = ((2 + 2) \bmod 3) + 1 = (4 \bmod 3) + 1 = 1 + 1 = 2$$

This means that the next available week is within the range of the cycle's **1st, 2nd Week and 3rd week** after the cycle of 3 weeks of scheduling given that **week 2 and 3 is occupied**.

Using Diophantine Equation to distribute patients

Let a, b, c be the number of patients assigned to Week 1, Week 2, and Week 3. The equation will be:

$$a + b + c \leq 120$$

To find the integer solution that each week does **not exceed 120 patients per cycle**.

One possible solution is:

$$a = 40, b = 40, c = 40 \mid a + b + c \leq 120$$

$$40 + 40 + 40 \leq 120$$

This means that:

- Week 1 : **40 patients will be scheduled**
- Week 2: **40 patients will be scheduled**
- Week 3 : **40 patients will be scheduled**

Over the schedule of one full 3 week cycle, all 120 patients are scheduled in a balanced way.

And the remaining 20 will be scheduled in the next cycle

Scenario:

A restaurant has **3 available weeks** for scheduling, **172 guests** need reservation. The restaurant can handle **no more than 40 guests** per week. Today is **Week 1**,

$$Gap = (total\ weeks + 1) - current\ week$$

$$Schedule = ((Current\ Week + Gap) \bmod Total\ Weeks) + 1$$

We have:

$$Gap = (total\ 3 + 1) - 1 = 3$$

$$Schedule = ((1 + 3) \bmod 3) + 1 = (4 \bmod 3) + 1 = 1 + 1 = 2$$

This means that the next available week is within the range of the cycle's **1st, 2nd Week and 3rd week** of the cycle of 3. In short, the possible schedule dates **from week 1 to 3rd week**.

Calculate the Total Weeks Needed

$$\frac{172}{40} = 4.3 = 5\ weeks$$

5 weeks are needed to serve all 172 guests.

Also,

$$172 = 40(4) + 12$$

Means that the remaining 12 guests are left for a partial week. Then, 40 + 12 guests will be left since it is a **3 week cycle**

Using Diophantine Equation to distribute patients

Let a, b, c be the number of patients assigned to Week 1, Week 2, and Week 3. The equation will be:

$$a + b + c \leq 120$$

Where,

$$a \leq 40$$

$$b \leq 40$$

$$c \leq 40$$

To find the integer solution that each week does **not exceed 40 guests per week and 120 per cycle**.

One possible solution is:

$$a \leq 40, b \leq 40, c \leq 40 \mid a + b + c \leq 120$$

$$40 + 40 + 40 \leq 120$$

This means that:

- Week 1 : **40 guests will be scheduled**
- Week 2: **40 guests will be scheduled**
- Week 3 : **40 guests will be scheduled**

Over the schedule of one full 3 week cycle, all 120 guests are scheduled in a balanced way.

And the remaining **52 (40 + 12) will be scheduled in the next cycle**.

To solve the remaining Guests, **another diophantine is required for solution.**

Then,

To find the integer solution that each week does **not exceed 40 guests per week and 120 per**
(for the remaining guests of the first calculation)

$$a + b + c \leq 120$$

One possible solution is:

$$40 + 12 + 0 \leq 120$$

This means that:

- Week 1 : **40 guests will be scheduled**
- Week 2: **12 guests will be scheduled (will be added when someone did a reservation)**
- Week 3 : **0 guests will be scheduled (will be added when someone did a reservation)**

Or to maintain the balance, one possible solution is:

$$18 + 17 + 17 \leq 120$$

- Week 1 : **18 guests will be scheduled**
- Week 2: **17 guests will be scheduled**
- Week 3 : **17 guests will be scheduled**

Analysis:

Clinic Scheduling:

Efficient scheduling is important not only for improving patient care but also to maintain the healthcare of the physicians. According to a 2018 survey (Rajae, 2022), physicians visit between 11 and 20 patients per day, with an average time of 17 to 24 minutes with each. This is equal to 77 to 140 patients every week. However, increasing the number of patients scheduled without efficient scheduling systems is very stressful to the physician, which has negative effects on diagnostic and overall healthcare quality that they provide.

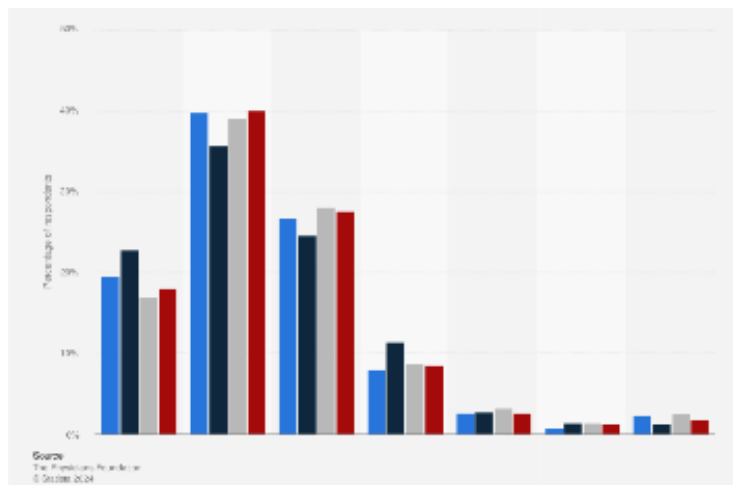


Fig 1 Number of patients that physicians in the US saw per day

(How Many Patients Does a Doctor Have a Day?)

Importance of the Application:

Unmanageable workloads are harmful to physicians' ability and also to their healthcare. Additionally, patients are at risk of receiving poor service or incorrect diagnosis. Furthermore, clinics must work within the limits of their available time constraints to ensure a proper

distribution of patients. When the demand surpasses weekly capacity, this can become very complicated. In these situations it requires mathematical solutions to maintain efficiency.

Mathematical Approach:

Given a clinic's weekly scheduling cycle and a certain number of patients requiring scheduling, it is important to maximize how these schedules are divided. With each week there are a certain number of patients that can be accommodated, and this scheduling system ensures that patients return every X week. To solve this, there are mathematical concepts that can be used such as **Modular Arithmetic and Diophantine Equations**. **Modular Arithmetic** helps to determine the continuous scheduling patterns, and **Diophantine Equations** helps in the distribution of patients properly over the range of weeks while staying within weeks limits. With this method, it assures that the scheduling system is efficient and effective.

Restaurant Reservation:

The main issue is how to efficiently manage a limited capacity restaurant in order to accommodate a possibly high number of guests over time. This requires long-term planning to identify the minimum number of week cycles needed, equal service distribution among available slots, and management of capacity usage to maximize income and the satisfaction of guests.

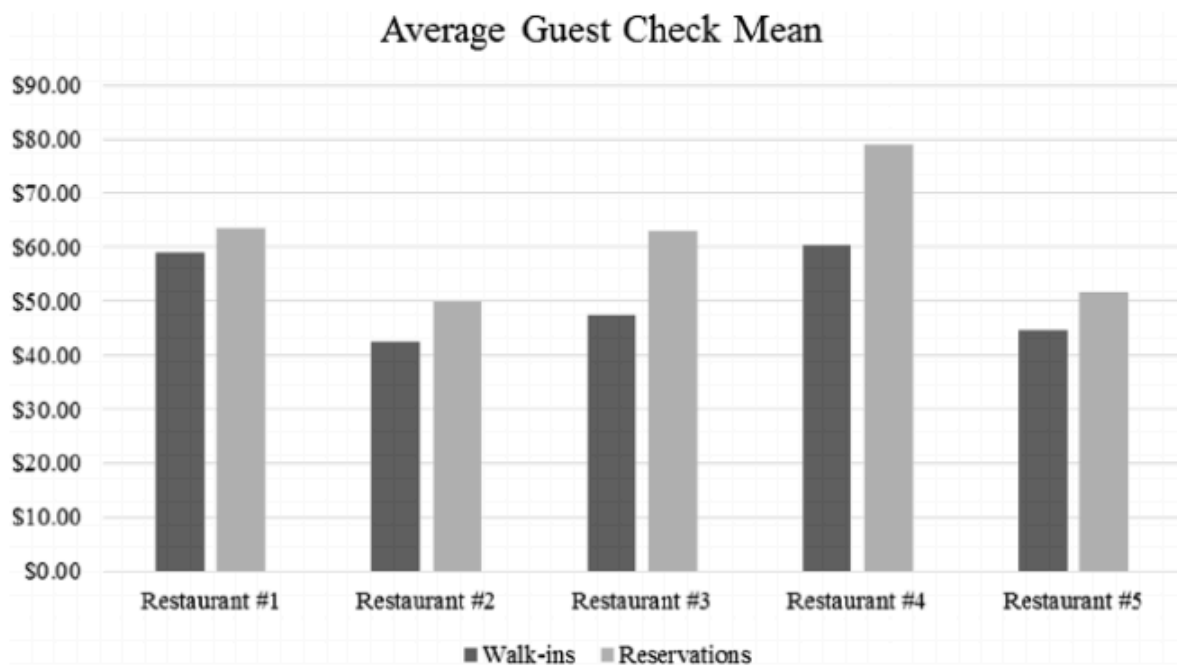


Fig 2 Average Guest Check Mean (Rosett, 2025b)

According to the reference gather an average size reservation only restaurant can accommodate 40 capacity in total, and from the data table the average reservation guest per month from these 5 restaurant is around 172 guests and in a 3 week reservation cycle

Table 1 Restaurant guest count and mean check totals

	Restaurant 1	Restaurant 2	Restaurant 3	Restaurant 4	Restaurant 5
Walk-in guests	157	152	169	43	35
Check mean in dollars	58.98	42.35	47.48	60.44	44.63
Reservation guests	172	113	167	266	134
Check mean in dollars	63.58	49.85	63.04	79.09	51.66
Variance	4.60	7.50*	15.56*	18.65*	7.03

* Denotes statistical significance

Fig 3 Restaurant guest count and mean check totals (Gregorash, 2016b)

Importance of the Application:

This problem also proves how important effective scheduling is in real-world situations. It helps with long-term planning by knowing how many cycles are required to handle every upcoming reservation. It also ensures equal service distribution through a fair and systematic way of distributing reservations to the available slots. Additionally, it improves capacity usage with an efficient management of their capacity to maximize income and customer satisfaction.

Mathematical Approach:

Given that a restaurant can only accommodate a certain number of guests because of the limited slot in the restaurant, then it is easier to solve this by using mathematical concepts. It starts by dividing the total guests with the number of capacity per week, this tells the number of weeks required to accommodate all guests. After this, using **Modular Arithmetic** and **Diophantine Equation** to get the week number within the cycle and to determine the number of guests for each week. **Modular Arithmetic** is responsible for getting the week's number. By calculating using the modulo operator, it is easy to understand a restaurant's availability in a week's cycle. On the other hand, **Diophantine Equation** helps to determine the suitable number of guests by using the formula $a + b + c \leq 120$, this will solve the number of guests assigned to each week within a 3-week cycle.

Conclusion

The Methods have proven to be effective in sorting appointments to provide smoother queues and better schedules. The effects not only saved time but also resulted in better quality of service where both the people doing the work and the ones requiring the service have gotten benefits from the results. Addressing the unbalanced workload has achieved great results and both in terms of efficiency and in maintaining a healthy working environment. This method is not limited to only these settings but can also be applied to things such as reservations for a place like resorts or hotels, it is also applicable to many different government related services, and even for services of things like catering and party related events. Overall it is a system designed to help organize and assist in making schedules.

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